

SHEENA PANTHAPLACKEL

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EDUCATION

The University of Texas at Austin, Austin, TX 2017 – 2022

Ph.D. in Computer Science

Advisors: Raymond Mooney, Junyi Jessy Li

Thesis: *Facilitating Software Evolution through Natural Language Comments and Dialogue*

University of Illinois at Urbana-Champaign, Champaign, IL 2013 – 2016

B.S. in Computer Science, Graduated with High Honors

EXPERIENCE

Google DeepMind, Mountain View, CA Oct 2022 – Present

Senior Research Scientist, *Gemini Post-Training for Agentic Coding*

- Design and implement strategies for creating and acquiring diverse, high-quality SFT and RL data focused on coding capabilities
- Run large-scale experiments to integrate new datasets into SFT and RL data mixtures
- Analyze usage patterns to identify key losses and build aligned evals for hill-climbing in post-training

Bloomberg L.P., New York, NY May 2021 – Aug 2021

AI Engineering Intern, *NLP Platform (Mentor: Adrian Benton)*

- Investigated approaches for representing news edits and generating updated headlines
- Built a model to automatically update news article summaries as stories evolved over time

The University of Texas at Austin, Austin, TX Jan 2019 – Aug 2020

Graduate Research Assistant, *Machine Learning Group*

- Conducted research at the intersection of machine learning, natural language processing, and software engineering
- Led a reading group on papers at the intersection of NLP and software engineering

Microsoft Research, Cambridge, UK Jun 2019 – Aug 2019

AI Research Intern, *Deep Program Understanding Project (Mentor: Miltos Allamanis)*

- Built machine learning models that comprehend and learn generalizable characteristics of source code edits
- Developed neural architectures for representing structural properties of code

The University of Texas at Austin, Austin, TX Aug 2017 – Dec 2018

Graduate Teaching Assistant, *Computer Science*

- CS 371R: Information Retrieval and Web Search (Fall 2018), CS 378: Declarative Programming (Spring 2018), CS 329E: Elements of Software Engineering (Fall 2017)

Spot Trading LLC, Chicago, IL Jul 2016 – Jul 2017

Software Developer

- Developed efficient message gateways to and from various exchanges
- Built infrastructure to rapidly process large volumes of real-time market data
- Designed trader-facing applications that leverage market data to identify profitable trading opportunities

Spot Trading LLC, Chicago, IL Jun 2015 – Aug 2015

Software Developer Intern

- Developed trading application features enabling traders and risk managers to model risk measures for future dates
- Restructured and optimized a compliance system that detects potential discrepancies and generates alerts based on real-time monitoring of the firm's market activity
- Investigated Redis-based performance optimizations for the system providing market data to applications across the firm

Motorola Solutions, Schaumburg, IL May 2014 – Aug 2014

IT Intern

- Designed a process to pull data from RSS feeds and display it on live tiles on the Windows 8.1 Start screen, adding tile functionality with JavaScript

University of Illinois at Urbana-Champaign, Champaign, IL Jan 2014 – May 2014

Undergraduate Researcher in Engineering

- Investigated how human presence in images influences people's reactions to those images in online social spaces

- Processed images using MATLAB's human detection tool and developed scripts for error analysis

University of Illinois at Urbana-Champaign, Champaign, IL Jan 2014 – Dec 2015

Course Assistant, Computer Science

- CS 241: System Programming (Spring 2015, Fall 2015), CS 125: Introduction to Computer Science (Spring 2014, Fall 2014)

PUBLICATIONS

- Gemini Team, Google. Gemini 2.5: Pushing the Frontier with Advanced Reasoning, Multimodality, Long Context, and Next Generation Agentic Capabilities. *arXiv preprint arXiv:2507.06261*, 2025.
- Miltiadis Allamanis, **Sheena Panthaplackel**, and Pengcheng Yin. Unsupervised Evaluation of Code LLMs with Round-Trip Correctness. *International Conference on Machine Learning (ICML)*, 2024.
- Sheena Panthaplackel**, Milos Gligoric, Junyi Jessy Li, and Raymond J. Mooney. Using Developer Discussions to Guide Fixing Bugs in Software. *Findings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2022.
- Sheena Panthaplackel**. Facilitating Software Evolution through Natural Language Comments and Dialogue. *Ph.D. Thesis, The University of Texas at Austin*, 2022.
- Jiyang Zhang, **Sheena Panthaplackel**, Pengyu Nie, Junyi Jessy Li, and Milos Gligoric. CoditT5: Pretraining for Source Code and Natural Language Editing. *IEEE/ACM International Conference on Automated Software Engineering (ASE)*, 2022.
- Sheena Panthaplackel**, Adrian Benton, and Mark Dredze. Updated Headline Generation: Creating Updated Summaries for Evolving News Stories. *Association for Computational Linguistics (ACL)*, 2022.
- Sheena Panthaplackel**, Junyi Jessy Li, Milos Gligoric, and Raymond J. Mooney. Learning to Describe Solutions for Bug Reports Based on Developer Discussions. *Findings of the Association for Computational Linguistics (ACL)*, 2022.
- Sheena Panthaplackel**, Miltiadis Allamanis, and Marc Brockschmidt. Copy That! Editing Sequences by Copying Spans. *The AAAI Conference on Artificial Intelligence (AAAI)*, 2021.
- Sheena Panthaplackel**, Junyi Jessy Li, Milos Gligoric, and Raymond J. Mooney. Deep Just-In-Time Inconsistency Detection Between Comments and Source Code. *The AAAI Conference on Artificial Intelligence (AAAI)*, 2021.
- Jiyang Zhang, **Sheena Panthaplackel**, Pengyu Nie, Junyi Jessy Li, Raymond J. Mooney, and Milos Gligoric. Leveraging Class Hierarchy for Code Comprehension. *NeurIPS Workshop on Computer-Aided Programming*, 2020.
- Sheena Panthaplackel**, Pengyu Nie, Milos Gligoric, Junyi Jessy Li, and Raymond J. Mooney. Learning to Update Natural Language Comments Based on Code Changes. *Association for Computational Linguistics (ACL)*, 2020.
- Sheena Panthaplackel**, Milos Gligoric, Raymond J. Mooney, and Junyi Jessy Li. Associating Natural Language Comment and Source Code Entities. *The AAAI Conference on Artificial Intelligence (AAAI)*, 2020.

HONORS & AWARDS

Bloomberg Data Science Ph.D. Fellowship 2020 – 2022

Edmund James Scholar 2014 – 2016

College of Engineering Dean's List 2013 – 2016

FMC Award of Excellence Scholarship 2014

PROFESSIONAL ACTIVITIES

- Invited Talk: *Harnessing the Coding Capabilities of LLMs*, CosmicAI Horizons Conference, Austin, TX 2025
- Reviewer, ICLR 2024, 2025
- Reviewer, NeurIPS 2023, 2025
- Reviewer, AAAI 2021, 2022, 2025
- Reviewer, COLM 2024
- Reviewer, ACL Rolling Review (ARR) 2023, 2024
- Reviewer, ACL 2023
- Reviewer, EMNLP 2021, 2023
- Reviewer, NLP4Prog Workshop at EMNLP 2021
- Reviewer, Mining Software Repositories Mining Challenge Track at ICSE 2021
- Reviewer, Student Research Workshop at AACL 2020
- Reviewer, Student Research Workshop at ACL 2020
- Reviewer, Student Research Workshop at NAACL 2019

SOCIETY MEMBERSHIPS

Association for the Advancement of Artificial Intelligence (AAAI) 2020 – Present

Association for Computational Linguistics (ACL) 2018 – Present

Society of Women Engineers (SWE) 2013 – Present

Women in Computer Science (WCS) 2013 – 2016